How do analyst recommendation changes affect subsequent firm-level stock prices in the technology industry?

Nick Kent

ECON 381

Macalester College

Abstract

This paper examines how analyst recommendation changes affect stock prices for publicly traded U.S. technology companies. Treating analyst recommendations as information events, this research uses a collapsed event-study framework to estimate the effect of analyst recommendations on cumulative abnormal returns (CARs). Using data on 36 technology firms and over 2,000 analyst recommendations from 2009 onward, abnormal returns are calculated using a market-adjusted model and summed over the [−5, 5] and [−10, 10] event windows. The regression results show that upgrades are associated with statistically significant increases in CARs, while downgrades generate negative CARs, with especially large effects for emerging firms. Overall, we conclude that analyst recommendation changes contain meaningful information.

Introduction

In the height of the COVID-19 pandemic Chinese factories were being hit hard and Apple was one of the first companies to be affected, and as a result its shares fell 25%. Goldman Sachs downgraded Apple while Daiwa Capital changed its evaluation from 'Buy' to 'Outperform'. By August 2020, Apple was trading at an all time high.

This paper will examine the relationship between analyst recommendation changes, upgrades or downgrades, and the corresponding change of a company's stock price in the tech industry. An upgrade occurs when an analyst raises their recommendation to a more favorable rating (e.g. 'hold' to 'buy'). On the flip side, a downgrade occurs when an analyst revises their recommendation to a less favorable rating (e.g. 'buy' to 'hold' or 'sell'). According to the Nasdaq, roughly 1,200 stock recommendations are made each day. These recommendations function as news events, and since analysts are seen to have superior information or models about company or industry fundamentals investors interpret as signals about a company's future performance.

When a company is upgraded the analyst is suggesting that the company has higher prospective growth or is improving its competitive positioning relative to other companies in the industry. This may cause investors to be excited about a company, increasing the demand for its stock and driving up the price. Furthermore, upgrades often reinforce optimism that investors already had, therefore amplifying existing optimism rather than creating it from scratch. On the contrary, when a company is downgraded by an analyst it might suggest that the company has weaker earnings potential or is overvalued. Investors might respond to this by selling the stock which in result drives the price of the stock down. It is reported that "the drifts following 'sell' recommendations are larger and last longer than those following 'buy' recommendations"

(Kudryavtsev, 73). This pattern likely reflects several underlying mechanisms, including greater investor sensitivity to negative information or cases where institutional investors might have to sell stocks that have unfavorable ratings.

This paper focuses on publicly traded technology firms and treats analyst recommendation changes as discrete information events. Using a collapsed event-study framework, it evaluates abnormal stock returns in short windows surrounding analyst upgrades and downgrades. This research will test the following two hypotheses. First, downgrades generate larger negative abnormal returns than the positive abnormal returns generated by upgrades. Second, sentiment-driven or immature firms (e.g., Palantir and Nvidia) experience larger abnormal returns following recommendation changes than mature firms (e.g., Apple and Oracle), with more negative reactions to downgrades and more positive reactions to upgrades.

Literature Review

There have been a handful of studies regarding the impact of analyst recommendations on stock prices. Analyst recommendations may reflect their ability to process public information or access private information, prompting people to either sell, hold, or purchase a company's stock (Ivkovich and Jegadeesh, 2004). Previous research finds that markets respond more strongly to negative recommendation revisions (e.g. ‘buy’ to ‘sell’) (Womack, 1996; Chang & Chan, 2008). This suggests that investors treat downgrades as more credible signals and shows clear risk aversion in investors (Chang & Chan, 2008). While most of these studies focused on the U.S. market, Murg, Pachler, and Zeitlberger found similar results in Austria, whereas in India buy recommendations generate short-term abnormal returns that diminish quickly and that sell recommendations do not significantly influence stock prices (Kumar, Chaturvedula, Rastogi, and Bang, 2009). Womack also found that downgrades exert greater influence in terms of the

persistence of stock price changes (Womack, 1996). The study found that stocks changed to a 'buy' recommendation saw a +3% average return in a three-day window, and +2.4% in the first month. However, this return then dissipated in the month to follow. Contrarily, stocks that were changed to a 'sell' recommendation saw a -4.7% average return over three days and -5.6% return over the first month to a -9.1% return over six months (Womack, 1996).

As discussed above, price drifts following downgrades tend to be larger than those following upgrades. With this, literature suggests that the direction and strength of these drifts depends on pre-event stock performance (Kurdryavtsev, 2021). Kudryavtsev (2021) found that if upgrades are preceded by low pre-event returns then the stock's price experiences stronger positive drifts, +1.5% return over six months (Kurdryasvtsev, 2021). If an upgrade is preceded by high pre-event returns it is common to see negative returns after the upgrade. Kudryavtsev found the same pattern with downgrades. If a downgrade is preceded by high pre-event returns, stocks show stronger negative drifts, -1.67% over six months (Kudryavtsev, 2021). If preceded by low pre-event returns, then downgrades are typically followed by price reversals. This proposes a potential trading strategy: to buy stocks that are upgraded after recent declines and short stocks that are downgraded after recent rises. Further work by Kudryavtsev found a relationship between the immediate reaction following recommendations and the long term outcome. Strong initial reactions predict reversals (suggesting an overreaction from the market), while weak ones predict gradual gains (Kudryavtsev, 2020).

The final theme to be discussed is the methodological consistency across all of these studies. The common approach is an event study framework which allows researchers to investigate the impact of an event (e.g. analyst recommendations) on stock prices over a defined time window (Brown & Warner, 1985). In this previous work the time frame was anywhere from

3 days (Loh & Stulz, 2009) all the way to 1-6 months post-event (Kudryavtsev, 2021; Womack, 1996). To quantify the effect of recommendations, the literature estimates cumulative abnormal returns (CARs). To do this, it would first be necessary to compute daily abnormal returns (ARs), defined as the difference between a stock's actual return and its expected return (typically estimated with the Market Model; Sharpe, 1963). Once ARs are obtained for each day in the window, they are summed over time to generate CARs, providing a measure of the total impact of the recommendation.

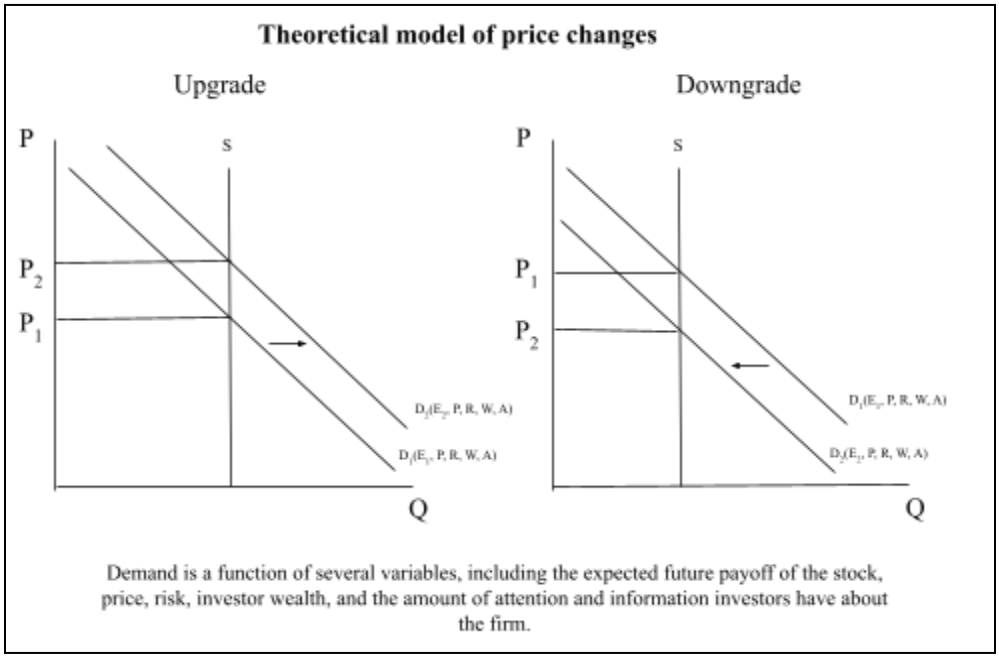
While lots of research has been done on analyst recommendations and stock prices at the market level, relatively little work has been focused on industry specific effects. This paper will address that gap by studying the technology sector, specifically comparing stock price reactions across firms that differ in maturity and sentiment-driven nature.

Theory

The relationship between analyst recommendations and stock returns can be understood using a simple supply and demand framework. In the short run, the supply is essentially fixed to the number of outstanding shares and does not change regardless of price, so the supply curve is a vertical line. The downward-sloping demand curve represents how many shares investors are willing to hold at different prices. Beyond analyst ratings or in other words expected future payoffs, investor demand for a stock depends on several other factors. These include risk, market conditions, investor wealth, and attention and information. When perceived risk rises without a corresponding increase in expected returns, investors become less willing to hold the stock, shifting the demand curve leftward. Market conditions, such as interest rates and economic uncertainty, affect how investors value future profits and therefore influence stock prices across the market. Investor wealth influences how much capital can be allocated to investments at any

given time. If an investor's wealth rises, demand for a company's stock may also rise as they look to expand their portfolios. Finally, differences in information and attention imply that not all investors observe or process news equally.

Analyst recommendations enter this model as they are a public signal that affects investors expected payoffs (Kraus & Stoll, 1972). In the case of an upgrade, perceived future payoffs and expected returns increase, meaning for any given price more investors want to hold the stock which shifts the demand curve to the right. With a fixed short run supply curve, the price must rise to clear the market. For a downgrade, analysts reveal information that lowers future expected returns meaning investors update their beliefs and want to sell their shares. We show this relationship in Figure 1.



Demand is a function of several variables, including the expected future payoff of the stock, price, risk, investor wealth, and the amount of attention and information investors have about the firm.

Figure 1

Lastly, we expect to see different reactions depending on the maturity of the firm. If a company is growing or in their early years, a recommendation change is a large information and attention shock which should cause price changes to be larger. Furthermore, emerging firms have

more uncertainty, less information history, and more volatile earnings which could also make analyst signals more influential. For larger or older companies the marginal signal from one analyst is smaller because fundamentals are already intensely scrutinized and incorporated into prices, therefore we expect price changes to be smaller.

Empirical Theory

We seek to estimate the effect that analyst recommendations have on short run stock price movements in the technology sector. To empirically test this causal relationship we use a collapsed event study framework that centers on the date of the analyst recommendation and measures the cumulative abnormal returns (CAR) in the days around the event. In a standard (dynamic) event-study framework, instead of collapsing the abnormal returns into a single value (cumulative abnormal returns) and estimating one coefficient, you estimate one coefficient per day. Although we are using a collapsed event-study framework in this study, future work might involve estimating dynamic betas which would allow us to test for pre-trends and timing of the effect.

The identification strategy for this analysis starts with isolating the effect of the analyst recommendation change from other factors that may influence stock price. To address this we start by only looking at the CAR in a short window, day -5 through day 5 relative to the recommendation date. We also include macroeconomic (federal funds rate and inflation rate) and firm level (EBITDA, profit, and assets) variables to reduce any confounding effects. These controls help ensure that the estimated effect of an analyst recommendation is not simply representing macro shocks or differences in firm fundamentals. Our preferred regression specification can be seen in Equation 1.

$$CAR_{it} = \beta_0 + \beta_1 RatingChange_{it} + \beta_2 Maturity_i + \beta_3 RatingChange_i * Maturity_i + \beta_4 FedFunds_t + \beta_5 InflationRate_t + \beta_6 AnnualAssets_{it} + \beta_7 AnnualEBITDA_{it} + \beta_8 AnnualProfit_{it} + \varepsilon_{it}$$

Equation 1

A potential concern with our identification strategy is if recommendation changes coincide with earnings announcements, product releases, or macroeconomic news, the CAR may reflect these events rather than the recommendation. The next issue that could arise is that if investors anticipate a recommendation change, based on prior price momentum or informal signals, the return effect may occur before the recommendation event. Finally, while we do include a few slow-moving macroeconomic and firm-level confounding variables, these may not be central to our identification strategy because event studies rely on short event windows to limit the influence of confounding variables.

Data Description

Data for this analysis was gathered from five sources. Daily stock price data (open, close, volume, etc) for 36 U.S. technology firms was obtained from Alpha Vantage's Daily Stock API. Analyst recommendation data (2009-present) on each of the 36 companies was gathered from Tradefeeds' Analyst Ratings API. For each recommendation there is information such as the analyst's firm, date of the rating, rating type (buy, hold, sell), and rating change (upgrade, downgrade, reiterated). The number of recommendations varies by firm, with some companies having over 150 recommendations and others fewer than 20. In total, the dataset contains 2,223 unique recommendations, and for each recommendation, stock price data was retained for 30 days before and after the recommendation. Then we gathered data on market returns from Yahoo Finance. This data contains the daily NASDAQ returns, which are used to control for overall market movements when analyzing firm-level price reactions. Annual financial fundamentals for

each company, including revenue, profit, assets, and EBITDA, were obtained from Macrotrends and aligned with the company's daily stock price data according to the year the stock price was from. Lastly, monthly CPI and interest rate data was gathered from FRED.

The dependent variable in this analysis is cumulative abnormal returns (CAR) measured over the event window, calculated by summing the firm's abnormal returns. Abnormal returns were calculated using a simplified version of the market model, the market-adjusted model, where abnormal return is equal to the security's return minus the market return. The primary independent variable is the analyst rating change. Control variables include firm-level financial fundamentals, macroeconomic indicators, and firm maturity.

Table 1 summarizes average firm-level and market-level (NASDAQ) log returns by rating change type. We can see that downgrades are associated with negative firm level returns and those returns are larger in magnitude than market-level returns. In contrast, upgrades correspond to positive firm-level returns that are once again larger than the market level returns on those same days. Table 2 presents summary statistics by rating type. Firms with a buy rating show small but positive average returns which are larger than market level returns, while those with hold or sell ratings exhibit negative average returns. Figure 2 illustrates the average normalized stock price movements surrounding analyst recommendations. We observe clear divergence in post-recommendation price dynamics, where stocks that are upgraded experience sustained positive price movements following the event, while downgraded stocks exhibit weaker performance. Additionally, in the pre-event window, prices associated with upgrades trend upward, whereas prices associated with downgrades trend downward. This pattern suggests that recommendation changes may partially reflect information already incorporated into prices prior to the announcement.

Summary Statistics by Rating Change Type

	Number of ratings	Mean	SD
Downgrades			
Firm-level log return	176	-0.0186	0.0512
Market (NASDAQ) log return	176	-0.0003	0.0107
Upgrades			
Firm-level log return	164	0.0215	0.0411
Market (NASDAQ) log return	164	0.0016	0.0126

Table 1

Summary Statistics by Rating Type

	Number of ratings	Mean	SD
Buy			
Firm-level log return	1291	0.0038	0.0539
Market (NASDAQ) log return	1291	0.0012	0.0124
Hold			
Firm-level log return	726	-0.0083	0.0892
Market (NASDAQ) log return	726	0.0013	0.0120
Sell			
Firm-level log return	154	-0.0102	0.0495
Market (NASDAQ) log return	154	0.0006	0.0107

Table 2

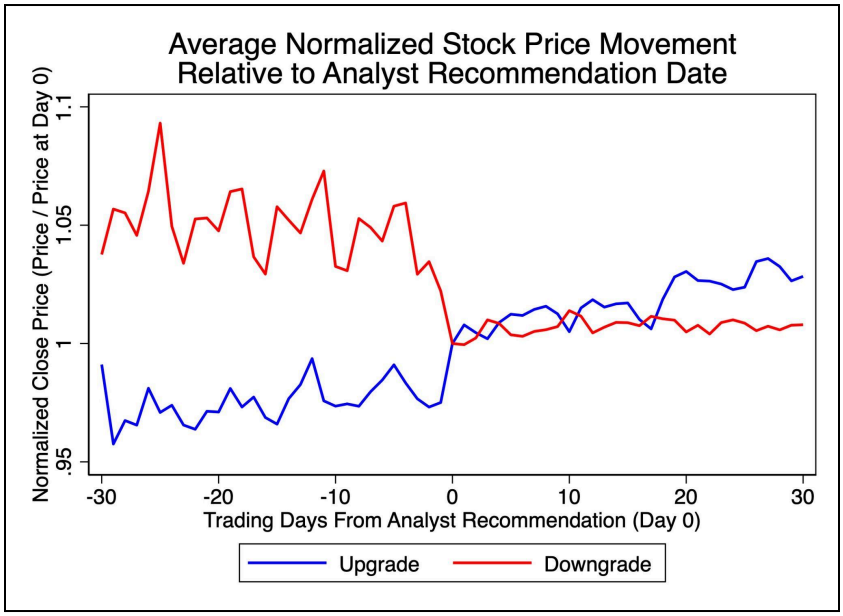


Figure 2

Results

The baseline regression in Table 3 Model 1 estimates the effect of rating changes on cumulative abnormal returns over the [-5, +5] event window, using maintained rating as the reference category. The constant term (-0.005) is not statistically significant, indicating that firms with unchanged ratings do not have abnormal returns that are different from zero over the 10 day event window. The downgrade coefficient tells us that stocks that receive a downgrade have an average CAR that is 2.3 percentage points lower than stocks that maintained the same rating. The upgrade coefficient tells us that stocks that receive an upgrade experience, on average, a 4.0 percentage point higher CAR over the event window compared to stocks that maintained the same rating. Both results are statistically significant at the 1% level.

Regression Estimates					
VARIABLES	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4	(5) Model 5
downgraded	-0.023***	-0.023***	-0.022***	-0.021**	-0.010
	(0.008)	(0.008)	(0.008)	(0.008)	(0.010)
upgraded	0.040***	0.040***	0.041***	0.042***	0.038***
	(0.008)	(0.008)	(0.009)	(0.009)	(0.010)
maturity_mid				0.008	0.009
				(0.008)	(0.009)
maturity_emerging				0.010	0.016
				(0.009)	(0.011)
downgraded # maturity_mid					-0.029
					(0.023)
downgraded # maturity_emerging					-0.051**
					(0.025)
upgraded # maturity_mid					0.029
					(0.027)
upgraded # maturity_emerging					0.007
					(0.023)
inflation_rate		-0.003**	-0.002	-0.002	-0.002
		(0.001)	(0.001)	(0.001)	(0.001)
fed_funds_rate		0.002	0.003*	0.003*	0.003**
		(0.001)	(0.002)	(0.002)	(0.002)
ln_assets			-0.009**	-0.007*	-0.007
			(0.004)	(0.004)	(0.004)
ln_ebitda			-0.005	-0.004	-0.004
			(0.004)	(0.004)	(0.004)
ln_profit			0.010	0.008	0.007
			(0.008)	(0.008)	(0.008)
Constant	-0.005	0.000	0.032	0.024	0.022
	(0.003)	(0.005)	(0.022)	(0.023)	(0.023)
Observations	1,238	1,238	1,171	1,171	1,171
R-squared	0.029	0.034	0.041	0.043	0.048

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 3: Regression Analysis Day -5 Through Day 5

Table 3 Model 2 extends the baseline specification by adding macroeconomic controls, the inflation rate and the federal funds rate. After controlling for these variables, upgrades and downgrades still have the same effect on CAR. Table 3 Model 3 includes control variables for firm size (assets) and profitability (EBITDA and profit). After adding these controls, the coefficients for upgrades and downgrades remain virtually unchanged. This tells us that price reactions to recommendation changes are not driven by firm size or profitability differences, but rather by the rating action itself. We then placed each firm in one of three maturity categories (mature[1], mid[2], or emerging[3]). Table 3 Model 4 shows that coefficients for maturity are small in magnitude and statistically insignificant, indicating no meaningful baseline differences in cumulative abnormal returns across maturity groups. Importantly, the estimates for upgrade and downgrade have remained highly stable. Downgrades continue to be associated with a statistically significant decline in CAR (-2.1 percentage points, p-value < 0.05), and upgrades remain associated with a large and statistically significant increase in CAR (4.2 percentage points, p-value < 0.01). These results show that firm maturity does not independently affect returns and does not confound the baseline effect of analyst recommendations. With that being said, maturity may still matter conditionally, meaning that firms of different maturity might respond differently to the same type of rating change.

Table 3 Model 5 introduces an interaction term that allows the effect of recommendation changes on cumulative abnormal returns to vary across maturity groups. The baseline maturity group is "mature", so all estimates are interpreted relative to mature firms. The downgrade coefficient is not statistically significant, which means that there is no reliable downgrade effect

[1] Mature firms. Firms classified as mature are well-established companies with stable operating histories, consistent profitability, and relatively predictable performances.
[2] Mid-maturity firms. Firms in the mid-maturity category exhibit characteristics between emerging and mature firms. They have moved beyond early-stage growth, but have not yet reached full financial stability.
[3] Emerging firms. Firms classified as emerging are early-stage or rapidly growing companies.

for mature firms. On the contrary, an upgrade raises CAR by an average of 3.7 percentage points (p-value < 0.01) for a mature firm, holding all other factors constant. The mid maturity and emerging maturity coefficients suggest that there is no baseline difference in CAR across maturity groups when ratings are held constant (rating change = maintained). The interaction between downgrade and mid-maturity is statistically insignificant which indicates that mid-maturity firms do not respond differently from mature firms to downgrades. The interaction between downgrades and emerging maturity is -0.051 and statistically significant at the 5% level. This tells us that downgrades trigger a reduction in CAR that is 5.1 percentage points greater for emerging firms than mature firms. This brings the total downgrade effect for emerging firms to a 6.1 percentage point decrease in CAR. The interaction between upgrade and mid-maturity is not significant meaning that upgrades do not affect mid-maturity and mature companies differently. Similarly, the interaction between upgrade and emerging maturity is not significant meaning that upgrades don't affect emerging firms differently than mature firms.

The above models were fit with CAR from a ten-day event window, day -5 through day 5. In Table 4 we widen the event window to 20 days, day -10 through day 10, and we get very similar results as Table 3 although the magnitude of the CAR estimates have grown. The downgrade coefficient in Model 5 remains statistically insignificant indicating that mature firms do not experience a change in CAR following a downgrade over a 20 day window. The interaction between downgrade and emerging maturity remains negative and statistically significant, while growing in magnitude compared to the baseline specification. Then, the upgrade coefficient continues to have a positive impact on CAR (4.4 percentage points, p-value < 0.01) and the interaction terms with maturity remain insignificant. One of the main differences between Model 5 in Table 3 and Table 4 is that in Table 4 the emerging maturity coefficient has

become statistically significant and larger in magnitude (2.9 percentage points, p-value < 0.05). This suggests that when a rating is maintained and all other control variables are held constant, emerging firms earn cumulative abnormal returns that are 2.9 percentage points higher than those of mature firms over the 20 day event window.

Regression Estimates					
VARIABLES	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4	(5) Model 5
downgraded	-0.029***	-0.029***	-0.027***	-0.026***	-0.013
	(0.010)	(0.010)	(0.010)	(0.010)	(0.012)
upgraded	0.042***	0.042***	0.044***	0.044***	0.044***
	(0.010)	(0.010)	(0.010)	(0.010)	(0.012)
maturity_mid				0.012	0.013
				(0.010)	(0.011)
maturity_emerging				0.017	0.029**
				(0.011)	(0.013)
downgraded # maturity_mid					-0.035
					(0.027)
downgraded # maturity_emerging					-0.057*
					(0.029)
upgraded # maturity_mid					0.039
					(0.033)
upgraded # maturity_emerging					-0.020
					(0.028)
inflation_rate		-0.006***	-0.004**	-0.005***	-0.005***
		(0.002)	(0.002)	(0.002)	(0.002)
fed_funds_rate		0.004**	0.005***	0.005***	0.005***
		(0.002)	(0.002)	(0.002)	(0.002)
ln_assets			-0.011**	-0.009*	-0.008
			(0.005)	(0.005)	(0.005)
ln_ebitda			-0.007	-0.005	-0.005
			(0.004)	(0.005)	(0.005)
ln_profit			0.013	0.010	0.010
			(0.009)	(0.010)	(0.010)
Constant	-0.004	0.005	0.049*	0.034	0.032
	(0.004)	(0.006)	(0.026)	(0.028)	(0.028)
Observations	1,238	1,238	1,171	1,171	1,171
R-squared	0.025	0.035	0.043	0.046	0.052

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 4: Regression Analysis Day -10 Through Day 10

Conclusion

Our results tell us that the baseline effect of an upgrade is associated with statistically significant positive change in cumulative abnormal returns. Over the 10 day event window there is a 3.8 percentage point increase in CAR and over the 20 day event window there is a 4.4

percentage point increase in CAR, holding all other factors constant. The interaction terms with maturity are insignificant regardless of the event window which suggests that favorable analyst ratings are incorporated similarly across firms regardless of their maturity. This partially contradicts our intuition as we assume that mature firms are already efficiently priced and therefore should not react to upgrades as strongly as mid-maturity or emerging firms.

On the contrary, the baseline effect of a downgrade is negative and statistically significant in both event windows but loses significance once maturity interactions are included in the full specification. This pattern suggests that the average downgrade effect is not the same across firms. In particular, the interaction between downgrades and emerging firms is negative and statistically significant in both Table 3 and Table 4, and it is larger in magnitude than the baseline downgrade coefficient. This indicates that emerging firms experience a larger decline in CAR than mature firms following a downgrade, this is consistent with our hypothesis. Interestingly, the interaction between downgrades and mid-maturity firms is statistically insignificant meaning that the larger negative reaction is not a part of a broader maturity gradient.

Lastly, the effects of downgrades and upgrades that are found in this paper are inconsistent with previous research, and do not support our hypothesis from earlier. In this paper, we found that the changes in CAR associated with an upgrade are more positive than the negative CAR changes associated with a downgrade. This contradicts previous research that claims that markets are more affected by negative recommendations than positive recommendations (Womack, 1996; Chang & Chan, 2008). With that being said, we did obtain results that are consistent with the idea that downgrades have more persistent effects (Womack, 1996). When we widened our event window from 10 days to 20 days the negative effect of

downgrades became more negative whereas the positive effect from upgrades did not increase by an equal amount.

References

Alpha Vantage Inc. (n.d.). Free stock APIs in JSON & Excel. https://www.alphavantage.co/

Barber, B. (2001). Can investors profit from the prophets? Security analyst recommendations and stock returns. Journal of Finance, 56(2), 531–563. https://www.proquest.com/scholarly-journals/can-investors-profit-prophets-security-analyst/docview/57011287/se-2

Brown, S. J., & Warner, J. B. (1985). Using daily stock returns: The case of event studies. Journal of Financial Economics, 14(1), 3–31. https://www.sciencedirect.com/science/article/abs/pii/0304405X8590042X

Chang, Y., & Chan, C. (2008). Financial analysts' stock recommendation revisions and stock price changes. Applied Financial Economics, 18(4–6), 309–325. https://www.proquest.com/scholarly-journals/financial-analysts-stock-recommendation-revisions/docview/56801311/se-2

Fama, E. F., Fisher, L., Jensen, M. C., & Roll, R. (1969). The adjustment of stock prices to new information. International Economic Review, 10(1), 1–21. https://doi.org/10.2307/2525569

Federal Reserve Bank of St. Louis. (n.d.). FRED economic data. https://fred.stlouisfed.org/

Investopedia. (2012, May 30). Understanding analyst ratings. https://www.investopedia.com/financial-edge/0512/understanding-analyst-ratings.aspx

Ivkovic, Z., & Jegadeesh, N. (2004). The timing and value of forecast and recommendation revisions. Journal of Financial Economics, 73(3), 433–463. https://doi.org/10.1016/j.jfineco.2004.03.002

Jegadeesh, N. (2004). Analyzing the analysts: When do recommendations add value? Journal of Finance, 59(3), 1083–1124. https://www.proquest.com/scholarly-journals/analyzing-analysts-when-do-recommendations-add/docview/56237560/se-2

Kraus, A., & Stoll, H. R. (1972). Price impacts of block trading on the New York Stock Exchange. Journal of Finance, 27(3), 569–588. https://doi.org/10.2307/2978262

Kudryavtsev, A. (2019). Effect of investor inattention on price drifts following analyst recommendation revisions. International Journal of Finance and Economics, 24(1), 348–360. https://www.proquest.com/scholarly-journals/effect-investor-inattention-on-price-drifts/docview/2376938980/se-2

Kudryavtsev, A. (2020). Stock return dynamics after analyst recommendation revisions. Journal of Risk and Control, 7(1), 1–16. https://www.proquest.com/scholarly-journals/stock-return-dynamics-after-analyst/docview/2508224540/se-2

Kudryavtsev, A. (2021). The correlation between stock returns before and after analyst recommendation revisions. Economic Annals, 66(228), 69–89. https://doi.org/10.2298/EKA2128069K

Loh, R. K., & Stulz, R. M. (2009). When are analyst recommendation changes influential? (NBER Working Paper No. 14971). National Bureau of Economic Research. https://doi.org/10.3386/w14971

MacroTrends LLC. (n.d.). MacroTrends historical data. https://www.macrotrends.net/

Murg, M., Pachler, M., & Zeitlberger, A. C. M. (2016). The impact of analyst recommendations on stock prices in Austria (2000–2014): Evidence from a small and thinly traded market. Central European Journal of Operations Research, 24(3), 595–616. https://doi.org/10.1007/s10100-014-0360-9

Muslu, V., Rebello, M., & Xu, Y. (2014). Sell-side analyst research and stock comovement. Journal of Accounting Research, 52(4), 911–954.

https://www.proquest.com/scholarly-journals/sell-side-analyst-research-stock-comovement/docview/1629323177/se-2

Nasdaq. (n.d.). Analyst recommendations and price targets (ARPT) [Data set]. Nasdaq Data Link. https://data.nasdaq.com/databases/ARPT

Sharpe, W. F. (1963). A simplified model for portfolio analysis. Management Science, 9(2), 277–293. https://doi.org/10.1287/mnsc.9.2.277

Tradefeeds. (n.d.). Analyst Ratings API. https://tradefeeds.com/analyst-ratings-api/

Womack, K. L. (1996). Do brokerage analysts' recommendations have investment value? Journal of Finance, 51(1), 137–167. https://doi.org/10.2307/2329305

Yahoo Finance. (n.d.). Yahoo Finance. https://finance.yahoo.com/

Yogesh Kumar, Chakrapani Chaturvedula, Nikhil Rastogi, & Nupur Pavan Bang. (2009). Impact of analyst recommendations on stock prices. The Icfai Journal of Applied Finance, 15(4), 39–52.